

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Built and deployed **AI-powered product features** using **Python, Flask, and scikit-learn** to automate transaction classification and anomaly detection across payment messaging systems, reducing manual review overhead by **40%** and enabling real-time alerting for **P1 incidents** in production environments serving millions of transactions daily.
- Architected end-to-end **data pipelines and ETL workflows** with **Python (pandas, NumPy)** and **SQL (PostgreSQL, Oracle, MySQL)** to consolidate **ISO 20022 message flows** and **SEPA payment data** into operational dashboards, ensuring **100% SLA compliance** and **audit-trail integrity** for regulatory and compliance reporting.
- Owned full production lifecycle from ideation to scaling—designed **Spring Boot REST APIs** for cross-border payment integration, deployed on **AWS (EC2, Lambda, CloudWatch)** with **Docker containerization**, and established **Jenkins CI/CD pipelines** that reduced deployment time by **60%** and enabled safe **change-control management** across interconnected financial systems.
- Simplified complex system architectures by refactoring monolithic **Java Spring Boot** services into microservices, improving code maintainability and reducing incident response time from **45 minutes to 12 minutes** through enhanced **Splunk and Dynatrace monitoring dashboards** and structured **Linux log analysis** workflows.
- Amplified team impact by integrating **AI tools into the development workflow**—leveraged machine learning for intelligent code review recommendations, test-case generation, and documentation automation, accelerating feature delivery by **35%** while maintaining high standards for correctness and reliability in regulated financial environments.
- Collaborated cross-functionally with product teams, business analysts, and compliance stakeholders to identify high-impact **AI transformation opportunities** in payments operations and customer experience, translating **SWIFT MT/MX message formats** and **PSD2 / Open Banking requirements** into robust, production-grade software solutions used by millions of end-users globally.

SKILLS

AI & Machine Learning – scikit-learn, supervised learning, unsupervised learning, anomaly detection, model evaluation, feature engineering, AI-powered product experiences.

Core Programming & Frameworks – Python, Java, Spring Boot, Flask, FastAPI, REST API development, microservices architecture, end-to-end ownership.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, containerization, deployment automation, scalable systems.

Data Engineering & Analytics – SQL (PostgreSQL, MySQL, Oracle), ETL pipelines, data modelling, data warehousing, pandas, NumPy, Power BI, Tableau, DAX, Power Query.

DevOps & Reliability – CI/CD (Jenkins), Git, Linux, shell scripting, production monitoring (Splunk, Dynatrace), incident management, change-control, log analysis.

Financial & Regulatory Domain – SWIFT messaging (MT/MX), ISO 20022 message flows, SEPA payments, cross-border payments, PSD2, Open Banking, GDPR, audit trails, SLA management.

Soft Skills & Methodology – reasoning ability, problem-solving from first principles, drive and initiative, team collaboration, Agile/Scrum, Jira, Confluence, cross-functional communication.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

AI-Driven Real-Time Payment Anomaly Detection & Recommendation Engine (HCL Technologies)

- Designed and deployed a **machine learning-powered anomaly detection system** using **scikit-learn (Isolation Forest, Local Outlier Factor)** integrated into a **Spring Boot microservice** architecture on **AWS Lambda** to identify fraudulent and operational anomalies in **ISO 20022 and SWIFT MT/MX message flows** with **96% precision** and **94% recall**.
- Built a robust **Python data pipeline** with **pandas and NumPy** to ingest, normalize, and feature-engineer **SEPA and cross-border payment transaction data** from **PostgreSQL and Oracle databases**, processing **2.5M+ transactions daily** with sub-second latency and **100% audit-trail compliance**.
- Architected **REST APIs** to expose anomaly scores and AI-driven recommendations to operations dashboards, reducing manual review time by **65%** and enabling **P1 incident detection** within **30 seconds** of suspicious activity, directly improving SLA compliance to **99.95%**.
- Implemented end-to-end **CI/CD pipeline** using **Jenkins, Git, and Docker** with automated unit and integration testing, enabling safe canary deployments and rollback mechanisms that eliminated **3 production incidents** over 18 months.
- Collaborated with product and compliance teams to translate **PSD2 / Open Banking regulatory requirements** into ML-driven detection rules, ensuring all AI recommendations included traceable audit trails and human-in-the-loop approval workflows for regulatory sign-off.

Multi-System Payments Operations Dashboard with AI-Powered Insights (HCL Technologies)

- Engineered an executive **Power BI dashboard** consolidating real-time **ISO 20022 message flow metrics, SEPA payment volumes, and cross-border reconciliation data** from 5 interconnected ERP and payment systems, delivering **10 key performance indicators** with **sub-minute refresh cycles** to senior management and regulatory teams.
- Applied **DAX calculations and Power Query ETL** to harmonize **PostgreSQL, MySQL, and Oracle data sources**, creating unified dimensional models that enabled automated regulatory reporting (GDPR, audit trails) and reduced monthly reconciliation effort by **12 hours**.
- Integrated AI-generated insights via Python-backed **FastAPI microservice** that scored payment corridors and customer segments by risk and profitability, driving a **18% improvement** in operational efficiency and enabling data-driven pricing decisions.
- Delivered interactive **Tableau visualizations** for PSD2 compliance monitoring and Open Banking transaction flows, enabling the operations team to identify regulatory exceptions within **2 hours** instead of **2 days**, directly supporting audit and compliance sign-offs.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)

- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)